Debugging Overdue Tasks Function

https://gemini.google.com/app/02ebbb9f6f4072b7

User prompt: task_manager.py: ```python from datetime import date def add_task(tasks, title, due_date=None, priority="medium"): """Add a new task to the task list. Args: tasks (list): The list of existing task dictionaries. title (str): The task's title. due_date (date, optional): The task's due date. Defaults to None. priority (str): One of "high", "medium", "low". Defaults to "medium". Returns: list: The updated list of tasks. """ tasks.append({"title": title, "due_date": due_date, "priority": priority, "completed": False}) return tasks def mark_complete(tasks, title): """Mark a task as completed by title. Args: tasks (list): The list of task dictionaries. title (str): The title of the task to mark complete. Returns: bool: True if a matching task was found and updated, False otherwise. """ for task in tasks: if task["title"] == title: task["completed"] = True return True return False def get_overdue_tasks(tasks, current_date): overdue = [] for task in tasks: if task["due_date"] < current_date and not task["completed"]: overdue.append(task) return overdue def calculate_completion_rate(tasks): if not tasks: return 0.0 completed = sum(1 for t in tasks if t["completed"]) return completed / len(tasks) ``` Debug get_overdue_tasks. It crashes with TypeError when a task has due_date=None. Give a concise step-by-step execution trace explaining the cause before the corrected function. Preserve the meaning: return incomplete tasks with a date strictly before current_date, and do not modify the input. Return only the short explanation and corrected function.

Response: Step-by-Step Execution Trace

1. **Invocation**: `get_overdue_tasks(tasks, current_date)` is called with a list containing a task where `due_date=None`.
2. **Iteration**: The `for` loop evaluates the first task (e.g., `task = {"title": "Task 1", "due_date": None, ...}`).
3. **Condition Evaluation**: Python evaluates `task["due_date"] < current_date`.
4. **Comparison**: Since `task["due_date"]` is `None`, Python attempts to evaluate `None < current_date`.
5. **Crash**: Python raises `TypeError: '<' not supported between instances of 'NoneType' and 'datetime.date'` because `None` cannot be compared to a `date` object using ordering operators.

Corrected Function

```
def get_overdue_tasks(tasks, current_date):
    overdue = []
    for task in tasks:
        if task["due_date"] is not None and task["due_date"] < current_date and not task["completed"]:
            overdue.append(task)
    return overdue
```
